Engineering Mechanics 1 · Unit_04 · UNIT 4

DRAFT — UE26ME142A · Unit_04 · UNIT 4 · CLASS 39 Moment of Inertia Numerical

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

CLASS 39 Moment of Inertia Numerical

1 CLASS 39 Moment of Inertia Numerical

Study focus: CLASS 39 Moment of Inertia_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000030 · EV0000031

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

The first question is A slash six Determine polar moments of inertia of a semicircular area about points A and B.

Key details

- What is polar moment of inertia?
- It's a moment of inertia about Z axis passing through that particular point.
- Since we are talking about polar moment of inertia about A, it is the moment of inertia about Z-axis passing through point A.
- And remember that z axis will be always normal to the plane of the paper.
- Ah the point A itself, that's the point to which the Z axis is passing, and that is normal to this plane of this semi-circular area.
- When we don't have a standard formula, when we don't have a standard formula, we have to apply the parallel axis theorem.
- Inertia about any axis is equal to inertia about parallels intoidal axis plus C area into distance square.
- This is the statement which we know.
- Since we are talking about the Z axis, here the centroidal axis also should be the centroidal Z axis.
- Next, where will be the centroidal?
- Centroidal Z axis will be passing through C and that will be normal to the plane of the paper.
- Remember the basic formulas whenever we have a semicircle.
- First locate the center.
- This is the center of a semicircle.

- Then draw the axis of symmetry.
- Then the distance.
- From the centroid to the center on the axis of symmetry is 4 r by 3 pi.
- This is 4 r by 3 pi, we will get this point C.
- That is nothing but z-axis passing through C.
- For that, remember I z will be equals to Ix plus Iy, since here it is the Z axis passing to the centroid, it will be a centroidal z axis.
- That will be equal to I bar X plus I bar.
- Y where I bar x is moment of inertia about centroidal x axis and i bar y is moment of inertia about centroidal y axis.
- Moment of inertia about a centroidal z axis.
- Then what is i bar x?
- I bar x we know from the formulas it is 0.11 into r to the power of 4.
- Then what is i bar y?
- That is pi r to the power of 4 by 8.
- And these two.
- We are going to get high bar Z.
- Remember, this is the moment of inertia about the centroidal Z axis.
- Once we know moment of inertia about centroidal z axis, we can find moment of inertia about z axis about any other point.
- Z axis is passing through point A, we have to find inertia about that Z axis.
- How to find that?
- That will be equal to inertia about centroidal Z axis, that is I bar Z plus.
- Area into distance square.
- This is the statement.
- Then what is I bar Z?
- We know moment of inertia about centroidal z axis is 0.503 R to the power of 4.
- Plus, area is area of a semicircle, that is pi r square by 2.
- Then what is D?
- D is the distance between the two parallel axes.
- Here, one axis is passing through A and one more axis is passing through C.
- Then how to find this distance AC?
- To find this distance A C, look at this.
- This is C which is at a distance of four r by three pi.
- Then we need to find this distance A C.
- This A C will be equal to under root of AO square plus C O square because C O A is a right angle triangle.
- Then what is A C?
- A C is equal to under root of R square.
- That is AO plus 4 R by 3 pi whole square.
- We will get the value of distance between the centroid and Z axis and the Z axis passing through A.
- And Area we know.
- Substitute the value of area, then the distance and I bar z in the.

- Equation 1 and simplify we are going to get the polar moment of inertia about the point A.
- That is 2.35 into R to the power of 4.
- This is nothing but moment of inertia about z axis passing through A.

Example or application from the lecture

- Now let's start with the first point, polar moment of inertia of this semicircular area, about a point A.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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Core explanation · 05:00–09:59

Next, we need to find moment of inertia, polar moment of inertia about point B.

Key details

- Inertia about polar moment of inertia about point B will be equals to moment of inertia about centroidal z axis, that is I bar Z plus area into distance square.
- I bar z we know from the previous case.
- This is 0.503 R to the power of 4, substitute that.
- Then area is area of the semicircular lamina that is pi r square by 2.
- D is the distance between the centroidal Z axis and the Z axis passing through B.
- Nothing but this C B distance.
- Then how to find this C B?
- C B is equals to year to year OB is R, R minus O C is 4 R by 3 pi.
- Once we know the distance, area, then I bar said, substitute these three in the equation 2.
- And we have to simplify this.
- Then we will get the polar moment of inertia about the point B.
- This nothing but moment of inertia of this semicircular area about the z-axis passing through B.
- That is 1.023 into r to the power of 4.
- This is the answer.
- Next question A slash 7 determine the moment of inertia of a quarter circular area about the y axis.
- Then we have to apply parallaxis theorem.
- Inertia about y axis is equals to initiate about centroidal y axis plus area into distance square.
- This is the statement.
- Then what is I bar y?
- For a quarter circle, we know from the formulas I bar y is inertia about centroidal y axis that is 0.055 into r to the power of 4.
- Remember the formula?
- Substitute that i bar y value.
- Since e r is a, it is going to be 0.055 a to the power of 4.
- Okay, next we have A.

- A is area of this quarter circle that is pi r square by 4.
- Since r is a, it is it will be pi a square by 4.
- D is the.
- Distance between the y axis and the centroidal y axis.
- For that, where will be the centroid with respect to the radial line?
- It will be at a distance of 4 r by 3 pi.
- But what is the distance here?
- Distance is this distance we want.
- That is the distance between y ax and y naught axis.
- Then how to find this distance?
- This distance will be equal to a by 2 plus a minus of 4r by 3 pi.
- Here R is A, so substitute R S A in this we will get the value of D.
- It is 1.07 double phi into a.
- Substitute the value of I bar y area then D in this equation one, then simplify this, we will get the value of I as 0.96368 to the power of 4.
- This is the moment of inertia of this quarter circle about the about y axis.
- This is the next question A slash 19.
- Determine moment of inertia of the shaded area about the X axis.
- Look at this shaded area.
- Shaded area is only this.
- We have to find inertia of this shaded area above x axis.
- First thing, do we have any standard formula for the shaded area?
- We don't have.
- Then how to get this shaded area?
- If you observe this figure carefully.
- This is a quarter circle of radius A in that this triangle has been cut or removed of base A by 2 and the height A.

Example or application from the lecture

- Let's consider that again we are going to apply the parallelaxis theorem.
- Y axis is given here, which is at a distance of A plus A by 2 with respect to this origin of this.

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Core explanation · 10:00–14:59

Next, how to find inertia of the shaded area?

Key details

- Moment of inertia of the shaded area about x axis will be equal to.
- Moment of inertia of this entire quarter circle about x axis minus in that what do we need to detect?
- Inertia of a triangle about the same x axis.

- I x is equal to moment of inertia of a quarter circular lamina about x axis minus moment of inertia of a triangle about the same x axis.
- Next, what is moment of inertia of this quarter circular area about x axis?
- This x-axis is coinciding with the radial line.
- Here r is a, so it will be pi a to the power of 4 by 16.
- We are done with the quarter circle.
- Next come to the triangle.
- What is the moment of inertia of a triangle about its base?
- The standard formula is BHQ by 12.
- Here what is base A by 2, and what is H A.
- Substitute in this formula, we are going to get the value of Ix.
- Radius is a and base of the triangle is A by 2, and the height of the triangle is A.
- Substituting this the above equation, it's going to be pi A to the power of 4 by 16 minus basis.
- A by 2, take that as it is.
- H is J 8 cube by 12.
- Simplify this, we will get the value of Ix is 0.154 a to the power of 4.
- Remember, this is the moment of inertia of a shaded area about the x axis.
- That's it for today's class.
- If you come acros any doubts, you can post your queries in the PESU forum.

Example or application from the lecture

- That's what is given here.
- That's what is given here.
- Try to solve the problems given in the assignment as well as in the question bank.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 39 Moment of Inertia_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/049_CLASS_39_Moment_of_Inertia__Numerical__b51a45df3dcf2238.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 39 Moment of Inertia_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/049_CLASS_39_Moment_of_Inertia__Numerical__b51a45df3dcf2238.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 39 Moment of Inertia_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/049_CLASS_39_Moment_of_Inertia__Numerical__b51a45df3dcf2238.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps.
CLASS 39 Moment of Inertia_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/049_CLASS_39_Moment_of_Inertia__Numerical__b51a45df3dcf2238.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md have no readable `.slides.md` text; their original page images are included.
3. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.

4. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
5. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
6. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
7. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
8. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00010	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0051. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00011	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0051. Sidecar variants compare...
SRC00012	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0051. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00013	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0051. Sidecar variants compare...

C Revision and retrieval practice